Hyeonsoo Jang

Researcher, Korea Automotive Technology Institute (KATECH) | hsjang2@katech.re.kr | ORCID 0000-0002-2476-991X

EDUCATION

Gwangju Institute of Science and Technology (GIST) — M.S. in Electrical Engineering and Computer Science **Mar. 2021 – Feb. 2023**

Thesis: Continual Learning for Simultaneous Localization and Mapping
Advisor: Prof. Moongu Jeon
GPA: 4.21 / 4.5

Tech University of Korea (TUK) — B.S. in Electronics Engineering **Mar. 2015 – Feb. 2021**

Thesis: Smart Fire Helmet
Advisor: Prof. Sukhyun Seo
GPA: 4.46 / 4.5 | Valedictorian | Rank: 1st / 1,218 (University)

WORK EXPERIENCE

Korea Automotive Technology Institute **Feb. 2023 – Present**

Data Platform R&D Center, Big Data & AIDV R&D Division, Autonomous Driving Technology R&D Laboratory, Researcher — Aug. 2025 – Present

Big Data & SW Technology Division, Autonomous Driving Technology R&D Laboratory, Researcher — Feb. 2023 – Jul. 2025

ADAS, Integrated Safety Research & Validation

RESEARCH PROJECTS

Integrated 5M+ Camera System with Real-time Perception and Recording **Oct. 2026 – Present**

Participating Researcher | Sponsor: Ministry of Trade, Industry and Resources

- Scenario-based vehicle data acquisition

Integrated Safety Control and Multi-Stage Seat Belt Control for Enhanced Occupant Safety with Active–Passive Integration **Apr. 2026 – Sep. 2026**

Participating Researcher | Sponsor: Ministry of Trade, Industry and Resources

- Developing prediction models for collision characteristics and crash pulses, and optimization strategies for real-time pre-crash safety applications
- Developing occupant motion monitoring and driver emergency behavior prediction methods for critical scenarios, including sudden braking and Autonomous Emergency Braking (AEB)
- Designing simulation and real-vehicle validation frameworks for integrated safety control systems encompassing ADAS, in-cabin monitoring, and multi-stage seat belt (MSB) control

Intelligent Heating Technology Using Multi-modal HMI Based on Nano-heating Film **Jul. 2025 – Present**

Participating Researcher | Sponsor: Ministry of Trade, Industry and Resources

- Synchronized acquisition of RGB, thermal, temperature, humidity and CAN data

Development of PMAPS System Validation Technology for 2025 KNCAP **Jun. 2025 – Oct. 2025**

Participating Researcher | Sponsor: Hyundai Motor Company

- Performed KNCAP PMAPS (Pedal Misapplication Prevention System) validation tests
- Configured high-precision sensors and operated vehicle data acquisition systems for black-box testing

Development of a real-vehicle-based pre-crash evaluation method for integrated safety **Feb. 2025 – Present**

Participating Researcher | Sponsor: Hyundai Motor Company

- Developed a real-vehicle data acquisition platform using a GV70 for pre-crash safety evaluation.
- Collected multi-modal data including camera, radar, INS (GPS/IMU), CAN-FD, and RGB/IR in-cabin sensing data
- Developed a ROS 2-based synchronized multi-sensor data acquisition system
- Conducted scenario-based pre-crash experiments in proving ground (PG) environments, including real-vehicle data acquisition and test equipment operation for repeatable crash-relevant safety scenarios

Automotive Application Software for SDV Electronic Architecture Verification **Jan. 2025 – Dec. 2025**

Participating Researcher | Sponsor: Ministry of Trade, Industry and Resources

Development of Integrated Controller and Software for Performing Heterogeneous **Sep. 2024 – Present**

Functions based on Multicore SoC

Participating Researcher | Sponsor: Ministry of Trade, Industry and Resources

- Defined ADAS performance evaluation specifications for multicore SoC-based integrated controllers, incorporating regulations, safety standards, and NCAP protocols
- Conducting ADAS performance evaluation and analysis using vehicle-based testing in a Proving Ground (PG) environment

Driving Data Acquisition Device and Driving Behavior DB for Mixed Traffic of Human-driven and Autonomous Vehicles **Nov. 2023 – Dec. 2024**

Participating Researcher | Sponsor: Ministry of Trade, Industry and Resources

Driving Data for Special Road Environments (Advanced) **Jul. 2023 – Dec. 2023**

Participating Researcher | Sponsor: National Information Society Agency (NIA)

- Guidelines on capture methods and procedures for unpaved road driving data

Development of next generation ETC system semiconductor with security function based on electrification **Apr. 2023 – Dec. 2025**

Participating Researcher | Sponsor: Ministry of Trade, Industry and Resources

- Verified Hardware Security Module (HSM) functionality for Electronic Toll Collection (ETC) systems
- Designed automated validation software and test development procedures (TDP)
- Validated cryptographic algorithms including AES-128 and SHA-256

Development of on-site control technology for road traffic network control in the event of an incident or disaster **Feb. 2023 – Dec. 2024**

Participating Researcher | Sponsor: Korean National Police Agency

- Developed traffic coordination strategies for mixed traffic environments involving autonomous and human-driven vehicles
- Developed software for integration with I2V connected validation environments
- Designed evaluation scenarios and test cases to verify the reliability of route coordination models
- Established imitation learning–based route guidance framework for non-autonomous vehicle behavior modeling and autonomous vehicle path planning

Development of MPC5.5(+) ADAS vehicle verification technology **Feb. 2023 – Jul. 2024**

Participating Researcher | Sponsor: LG Electronics

- Designed and executed ADAS validation based on global safety regulations and OEM requirements (NCAP, ISO, UNECE)
- Operated high-precision measurement and in-vehicle sensor instrumentation systems for black-box testing of production ADAS vehicles under proving ground (PG) conditions
- Performed system-level evaluation of ADAS performance in PG environments and established structured validation procedures, including test execution and verification report generation

Development of AI-based robot and service content to support children's reading activities **May 2022 – Dec. 2022**

Participating Researcher | Sponsor: Ministry of Culture, Sports and Tourism

- Contributed to the design of the system framework

PUBLICATIONS

SCIE

[P01] (SCIE) Park, J., **Jang, H.**, Kang, B., and Hwang, Y., "Plausible Vehicle Trajectory Augmentation with Hybrid Machine Learning," IEEE Access, vol. 13, May 2025.

International Conferences

[P02] Kwag, S., Kang, B., **Jang, H.**, Park, J., and Hwang, Y., "Control Strategy for Cooperative Driving Automation in Traffic Incidents," IEEE ICCAD, pp. 1–6, May 2023.

Domestic Journals

[P03] (KCI) Hussain, M. I., Rafique, M. A., **Jang, H.**, and Jeon, M., "Camera–Radar Extrinsic Calibration for Autonomous Driving," KIISE Transactions on Computing Practices, vol. 29, no. 1, pp. 38–43, Jan. 2023.

Domestic Conferences

[P04] Kang, B., **Jang, H.**, Park, J., Kwag, S., and Hwang, Y., "ADAS Edge Case Detection Framework: Establishing Risk Thresholds and Application to Real-World AEB Cases," KSAE Fall Conference, Nov. 2025. **[Best Paper Award]**

[P05] Park, J., **Jang, H.**, Kang, B., and Kwag, S., “Extended Test Framework for Fault Tolerance Verification of CAN-FD Based ETC System SoC,” KSAE Fall Conference, Nov. 2025.

[P06] Kang, B., **Jang, H.**, Park, J., and Kwag, S., “Requirements Analysis for Evaluation of Electrified ETC System-on-Chip,” KSAE Fall Conference, Nov. 2024.

[P07] **Jang, H.**, Park, J., Kang, B., and Kwag, S., “Autonomous Vehicle Traffic Coordination Performance Evaluation in Intersection Incident Scenarios,” KSAE Fall Conference, Nov. 2024.

[P08] **Jang, H.**, Park, J., Kang, B., Hwang, Y., and Kwag, S., “Traffic Rules for Reinforcement Learning-Based Autonomous Driving in Incident Scenarios,” KSAE Fall Conference, Nov. 2023.

[P09] Jeong, S., Hong, Y., Hong, S., Hwang, Y., **Jang, H.**, Kwag, S., and Kang, B., “Multi-agent System for Autonomous Driving Assist in Traffic Incidents,” KSAE Fall Conference, p. 1615, Nov. 2023.

[P10] Park, J., **Jang, H.**, Kang, B., and Kwag, S., “Data Construction for Autonomous Vehicle Response in Intersection Incident Regions,” KSAE Fall Conference, Nov. 2023.

[P11] Hussain, M. I., Rafique, M. A., **Jang, H.**, and Jeon, M., “Camera–Radar Extrinsic Calibration for Autonomous Driving,” Proceedings of KSC (Korea Software Congress), Dec. 2021. **[Best Presentation Paper Award (Information & Communications)]**

[P12] **Jang, H.**, Kim, H., Song, J., Lee, J., and Seo, S., “Development of Smart Safety Fire-Fighting Helmet Based on Augmented Reality,” KICS Fall Conference, Nov. 2020. **[Encouragement Award, Undergraduate Paper]**

[P13] **Jang, H.**, Song, J., Kim, H., and Seo, S., “Perception Filter Design and Speed Control Method for Autonomous Driving Safety Improvement,” KSAE Fall Conference, Nov. 2020.

PATENTS

[T01] **Jang, H.**, Kang, B., Park, J., Kwag, S., and Lee, W., “Occupant Restraint Control Apparatus and Method,” Korean Patent Application (Filed, unpublished), 2026.

[T02] Kang, B., Park, J., **Jang, H.**, and Kwag, S., “Apparatus and Method for Verifying Communication Performance of Semiconductor Device,” Korean Patent Application (Filed, unpublished), 2025.

[T03] **Jang, H.**, Park, J., Kang, B., and Kwag, S., “Apparatus for Generating Avoidance Path in Autonomous Vehicle and Method Thereof,” Korean Patent Application No. 10-2024-0161193 (Filed), 2024; Publication No. 10-2026-0072231 (Published), 2026.

[T04] Park, J., **Jang, H.**, Kwag, S., and Kang, B., “Apparatus for Generating Traffic Flow Learning Data and Method Thereof,” PCT Application No. PCT/KR2024/010723 (Filed), 2024.

[T05] Park, J., **Jang, H.**, Kwag, S., and Kang, B., “Apparatus for Generating Traffic Flow Learning Data and Method Thereof,” Korean Patent No. 10-2678406 (Granted), 2024.

HONORS & AWARDS

[A01] Outstanding Award, Wise Research Life Idea Contest, GIST	Sep. 2021
[A02] Valedictorian Award, Ranked 1st among 1,218 undergraduate students, Gyeonggi Province	Feb. 2021
[A03] Finalist, 2020 Hanium ICT Contest, The Federation of Korean Information Industries	Dec. 2020
[A04] Grand Prize, Extracurricular Activity Essay Contest, Tech University of Korea	Nov. 2020
[A05] Encouragement Award, Industry–Academia Collaborative Project Presentation, Tech University of Korea	Nov. 2020
[A06] President's Award, 20th Industry–University Cooperation Technology Exhibition, Tech University of Korea	Oct. 2020
[A07] First Prize (1st place), CP-CoP Creative Project Competition, Tech University of Korea	Jul. 2020
[A08] Encouragement Award, CP-CoP Creative Project Competition, Tech University of Korea	Jul. 2019
[A09] Outstanding Award (1st place), Field Internship Essay Contest, Tech University of Korea	Jul. 2019

SCHOLARSHIPS

[S01] GIST Research Assistant Scholarship, GIST	Mar. 2021
[S02] Government Support Scholarship (M.S. program, selected at admission), GIST	Feb. 2021
[S03] Academic Achievement Scholarship A, Tech University of Korea	Sep. 2020
[S04] T.I.P Point Scholarship, Tech University of Korea	Sep. 2020
[S05] Department Specialization Scholarship, Tech University of Korea	Sep. 2020
[S06] Language Excellence Scholarship, Tech University of Korea	Sep. 2020
[S07] COVID-19 Special Scholarship, Tech University of Korea	Sep. 2020
[S08] IBK Scholarship for Academic Excellence, Industrial Bank of Korea	May 2020

[S09] President's Scholarship, Tech University of Korea — Mar. 2020
[S10] Dream Support Scholarship, Tech University of Korea — Sep. 2019
[S11] National Scholarship for Science and Engineering, Ministry of Science and ICT — Aug. 2019
[S12] Field Education Incentive, Tech University of Korea — Sep. 2018

TRAINING & CERTIFICATES

Getting Started with Research Tool Development through Vibe Coding, Hyundai NGV — Sep. 2026
Seminar on Next-Generation VLA-based End-to-End Autonomous Driving Strategies with Alpamayo — Jul. 2026
BMW Starter Pack Program, BMW — Jun. 2025
Vector CANoe Workshop, Vector Korea — Apr. 2025
Autonomous Driving Verification, Evaluation and Implementation Training — Mar. 2025

SKILLS & LANGUAGES

Programming: Python, C, C++, MATLAB
Languages: TEPS 410 / 600 (Jun. 2026)